Quantum Field Theory and the Standard Model – Matthew D. Schwartz

Notes by Gene

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Part I

Field Theory

1 Preliminary Details

1.1 $\hbar = c = 1$

Physicists like to set $\hbar = c = 1$ to 1) simplify equations, 2) reveal some symmetries. This makes all quantities some power of mass.

Note that:

$$c = 1 : [\text{Length}] = [\text{Time}]$$
$$\hbar = 1 : [\text{Time}]^{-1} = [\text{Energy}] = [\text{Mass}]$$

- Quantities with positive mass dimensions are "Energies."
- Quantities with negative mass dimensions are "Lengths."

Then further note:

$$[dx] = [x] = [t] = -1$$
$$[\partial_\mu] = [p_\mu] = 1$$

You'll get used to it.

1.2 Lorentz Transformations

Lorentz transformations are composed of rotations + boosts. They're generated by the **Lorentz group.** They preserve the Minkowski inner product:

$$\boxed{\Lambda^T g \Lambda = g}$$

i.e. $V_\mu W^\mu = V_0 W_0 - V_1 W_1 - V_2 W_2 - V_3 W_3$ is invariant.

- Note that we use $g_{\mu\nu} = \text{diag}(+1, -1, -1, -1)$ here.
- And that $g_{\mu\nu} = \eta_{\mu\nu}$ outside of exotic quantum gravity stuff.

$$\text{Timelike} : V^\mu V_\mu > 0$$
$$\text{Spacelike} : V^\mu V_\mu < 0$$
$$\text{Lightlike/Null} : V^\mu V_\mu = 0$$

2 Historic Background

2.1 Blackbody Radiation

Blackbody: an object at some temperature (internal structure is irrelevant)

- A box with size L has EM waves s.t.

$$\lambda = L/n \ , \quad v = f\lambda$$
$$\omega_n = \frac{2\pi}{L}|\vec{n}|c$$

- **Classical equipartition theorem:** light is emitted equally in all modes. The intensity is proportional to the differential volume of phase space, thus by dimensional analysis,

$$I(\omega) \equiv \frac{1}{V}\frac{d}{d\omega}E(\omega) \ \longrightarrow \ I(\omega) = \left[c^{-3}\omega^2 k_B T\right]$$

We expect $I(\omega)$ to grow with angular frequency (ω), but this experimentally the graph is a Maxwell-Boltzmann distribution. Called the "ultraviolet catastrophe."

Planck's solution: Energy in each EM mode is quantized.

$$\boxed{E_n = \hbar\omega_n}$$

- Interpretation: light is made up of massless particles ("photons")
- If we calculate $I(\omega)$ using this, the catastrophe is averted.
- The derivation (see book) uses equilibrium stat mech – so light must "equilibriate," i.e. particles must be created & destroyed. This is the domain of QFT.

$$I(\omega) = \frac{1}{V}\frac{dE(\omega)}{d\omega} = \frac{\hbar}{\pi^2}\frac{\omega^3}{e^{\hbar\omega\beta} - 1}$$

2.2 Second Quantization

2.2.1 Einstein Coefficients

Assume an atom with energy levels E_1, E_2 such that $E_2 - E_1 = \hbar\omega$. Further assume n_2 of the E_2 atoms and n_1 of the E_1 atoms.

A : Coefficient of spontaneous emission
B : Coefficient of stimulated emission
B' : Coefficient of absorption

A is the probability for a E_2 atom to become a E_1 atom by emitting a $\hbar\omega$ photon. B is the probability for an incoming $\hbar\omega$ photon to induce $E_2 \to E_1$. B' is the probability for an incoming photon to induce $E_1 \to E_2$.

We can estimate the differential dn_2 based on the above. Further, since particle number is conserved in this system,

$$dn_2 \approx -\left[A + BI(\omega)\right] n_2$$
$$dn_1 = -dn_2 = -\left[A + BI(\omega)\right] n_2 + B'I(\omega)n_1$$

Using equilibrium stat mech, Einstein was able to predict the relation between A, B, and B'.

2.2.2 Dirac

Recall the quantum harmonic oscillator with creation/destruction operators $a^\dagger$, a.

$$[a, a^\dagger] = 1$$
$$\hat{N} \equiv a^\dagger a \ , \quad \hat{N} \left|n\right\rangle = n \left|n\right\rangle$$
$$a^\dagger \left|n\right\rangle = \sqrt{n+1} \left|n+1\right\rangle$$
$$a \left|n\right\rangle = \sqrt{n} \left|n-1\right\rangle$$

(Be able to derive above) Now we use Fermi's golden rule to calculate the transition rates.

$$\Gamma \sim |\mathcal{M}|^2 \delta(E_f - E_i) \ , \quad \mathcal{M} = \langle f | H_{\text{int}} | i \rangle$$

H_{int} must have creation/destruction operators to create photons. It must also be Hermitian. Thus $H_{\text{int}} = H_I^\dagger a^\dagger + H_I a$. For the $2 \to 1$ transition, $|i\rangle = |\text{atom}_2\rangle \otimes |n_\omega\rangle$ and $|f\rangle = |\text{atom}_1\rangle \otimes |n_\omega + 1\rangle$ where n_ω (number of photons) has increased by one.

$$\begin{aligned}
\mathcal{M}_{2\to1} &= \langle f | H_{\text{int}} | i \rangle \\
&= \langle \text{atom}_1 | H_I^\dagger | \text{atom}_2 \rangle \langle n_\omega + 1 | a^\dagger | n_\omega \rangle + \langle \text{atom}_1 | H_I | \text{atom}_2 \rangle \langle n_\omega + 1 | a | n_\omega \rangle \\
&= \mathcal{M}_0^\dagger \sqrt{n_\omega + 1} \ , \quad \mathcal{M}_0^\dagger \equiv \langle \text{atom}_1 | H_I^\dagger | \text{atom}_2 \rangle
\end{aligned}$$

The $1 \to 2$ transition is very similar.

$$\mathcal{M}_{1\to2} = \mathcal{M}_0 \sqrt{n_\omega} \ , \quad \mathcal{M}_0 \equiv \langle \text{atom}_2 | H_I | \text{atom}_1 \rangle$$

Thus n_2 and n_1 change as

$$\begin{aligned}
dn_2 = -dn_1 &= -|\mathcal{M}_{2\to1}|^2 n_2 + |\mathcal{M}_{1\to2}|^2 n_1 \\
&= -|\mathcal{M}_0|^2 (n_\omega + 1) n_2 + |\mathcal{M}_0|^2 (n_\omega) n_1
\end{aligned}$$

Now we start the heavy calculations.

1. Energy is quantized by $\Delta = \hbar\omega = \hbar \frac{2\pi}{L} |\vec{n}|$ (with $c = 1$)

$$E(\omega) = \int^\omega (\hbar\omega) n_\omega \, d^3\vec{n} = (4\pi) \hbar L^3 \int_0^\omega \frac{d\omega}{(2\pi)^3} \omega^3 n_\omega$$

2. Multiply above by 2 for the 2 polarizations of light.

3. Intensity is

$$I(\omega) = \frac{1}{L^3} \frac{dE}{d\omega} = \frac{\hbar\omega^3}{\pi^2} n_\omega$$

So now we have

$$\boxed{dn_2 = -dn_1 = -|\mathcal{M}_0|^2 \left[1 + \frac{\pi^2}{\hbar\omega^3} I(\omega)\right] + |\mathcal{M}_0|^2 \left[\frac{\pi^2}{\hbar\omega^3} I(\omega)\right] n_1}$$

By comparison with the previous section, we read off the Einstein coefficients:

$$\boxed{B' = B\ , \quad \frac{A}{B} = \frac{\hbar}{\pi^2}\omega^3}$$

2.2.3 Key Ideas

Technically only one "quantization" is going on.

First quantization: Interpret modes as having quantized energy $E = \hbar\omega$
Use Hilbert space $\mathcal{H}_n$ for n-particle state
Second quantization: Quantize each mode as a harmonic oscillator
Use Fock space $\mathcal{F} = \oplus_n \mathcal{H}_n$ over all $0 \ldots n$-particle states

2.3 Harmonic Oscillators

2.3.1 Lorentz Invariance and Special Relativity

A simple Lorentz-invariant equation is

$$\Box\phi = (\partial_t^2 - \nabla^2)\phi = 0$$

An example solution is $\phi(x) = a_p(t)e^{i\vec{p}\cdot\vec{x}}$ with $(\partial_t^2 + \vec{p}\cdot\vec{p})a_p(t) = 0$. This is very similar to the harmonic oscillator equation $m\ddot{x} + kx = 0$.

The **general solution** sums over all p modes.

$$\phi(\vec{x}, t) = \int \frac{d^3p}{(2\pi)^3} \left[a_p(t)e^{i\vec{p}\cdot\vec{x}} + a_p^*(t)e^{-i\vec{p}\cdot\vec{x}}\right]$$

$$\text{Or: } \phi(\vec{x}, t) = \int \frac{d^3p}{(2\pi)^3} \left(a_p e^{-ipx} + a_p^* e^{ipx}\right)\ , \quad p_\mu \equiv (\omega_p, \vec{p}) \text{ and } px = p_\mu x^\mu$$

- This shows the decomposition of a scalar field into plane waves.
- Some vector fields can be dealt with this as well. Electromagnetic waves, for example, obey $\Box A_\nu = 0$ in each of the A_ν of the vector potential, given the Lorenz gauge $\partial_\mu A_\mu = 0$.

2.3.2 In Quantum Field Theory

We generalize the harmonic oscillator of quantum mechanics, $[a, a^\dagger] = 1$, to $[a_k, a_p^\dagger] = (2\pi)^3\delta^3(\vec{p} - \vec{k})$ with factors for convention.

As expected (with factors for convention),

$$a_p^\dagger |0\rangle = \frac{1}{\sqrt{2\omega_p}} |\vec{p}\rangle$$

$$\langle 0| a_p = \frac{1}{\sqrt{2\omega_p}} \langle\vec{p}|$$

Quantum fields are momentum integrals over these creation/annihilation operators for multiple oscillators.

$$\boxed{\phi_0(\vec{x}) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2\omega_p}}\left(a_p e^{ipx} + a_p^\dagger e^{-ipx}\right) \ , \quad px \equiv p_\mu x^\mu}$$

- By acting on $|0\rangle$ and projecting onto the momentum space, we see that

$$\langle\vec{p}|\phi_0(x)|0\rangle = e^{-ipx}$$

- I.e. the action of $\phi_0(x)$ is to create particles at position $\vec{x}$.

2.3.3 Time Dependence

In QFT, we often use the Heisenberg picture: time dependence is encoded in the operators.

$$a_p(t) = e^{-i\omega_p t}a_p$$
$$a_p^\dagger(t) = e^{i\omega_p t}a_p^\dagger$$

Thus we could **define** quantum scalar fields as

$$\boxed{\phi_0(\vec{x},t) = \int \frac{d^3p}{(2\pi)^3}\frac{1}{\sqrt{2\omega_p}}\left(a_p e^{-ipx} + a_p^\dagger e^{ipx}\right)}$$

3 Classical Field Theory

3.1 Hamiltonians and Lagrangians

Hamiltonians $\mathcal{H}[\phi,\pi]$ and **Lagrangians** $\mathcal{L}[\phi,\dot{\phi}]$ are functionals of fields + conjugate momenta.

$$H = \int \mathcal{H}\, d^3x \ , \quad L = \int \mathcal{L}\, d^3x$$
$$\mathcal{H}[\phi,\pi[\phi,\dot{\phi}]] = \pi[\phi,\dot{\phi}]\dot{\phi} - \mathcal{L}[\phi,\dot{\phi}]$$
$$\pi = \frac{\partial\mathcal{L}[\phi,\dot{\phi}]}{\partial\dot{\phi}}$$

So e.g. the general Lagrangian:

$$\begin{aligned}\mathcal{L} &= \frac{1}{2}(\partial_\mu\phi)(\partial_\mu\phi) - \mathcal{V}[\phi] \\ &= \underbrace{\frac{1}{2}\dot{\phi}^2}_{\mathcal{K}} - \underbrace{\frac{1}{2}(\nabla\phi)^2 - \mathcal{V}[\phi]}_{\mathcal{V}}\end{aligned}$$

Has the corresponding Hamiltonian:

$$\mathcal{H} = \pi\dot{\phi} - \mathcal{L}[\phi,\dot{\phi}] \ , \quad \pi = \frac{\partial\mathcal{L}}{\partial\dot{\phi}} = \dot{\phi}$$
$$\mathcal{H} = \underbrace{\frac{1}{2}\dot{\phi}^2}_{\mathcal{K}} + \underbrace{\frac{1}{2}(\nabla\phi)^2 + \mathcal{V}[\phi]}_{\mathcal{V}}$$

- **Kinetic terms** are marked by $\mathcal{K}$ above. They involve 2 fields (same or different). They tell us free-propagating behavior.
- If a field has time derivatives in its kinetic term, it's **dynamical.**
- Interaction terms are marked by $\mathcal{V}$ above. They involve 3+ fields.

Some examples of notation:

- Scalar fields: ϕ, π
- Fermions: ψ, ξ, χ
- Vector fields: A_μ, J_μ, V_μ
- Tensors: $h_{\mu\nu}$, $T_{\mu\nu}$

3.2 Noether's Theorem

3.2.1 Euler-Lagrange Equation

Start from the action S and vary it. After some reorganizing with the product rule,

$$S = \int d^4x \; \mathcal{L}(x)$$

$$\delta S = \int d^4x \left\{ \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right] \delta\phi + \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial((\partial_\mu \phi))} \delta\phi \right] \right\}$$

Assume boundary conditions such that ϕ vanishes at ∞. We can drop the last term and recover the **Euler-Lagrange equation** of motion.

$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) = 0}$$

3.2.2 Symmetries

If a lagrangian is invariant under some transformation, that transformation is called a **symmetry** of the Lagrangian. E.g. $\phi \to e^{-i\alpha}\phi$ for $\alpha \in \mathbb{R}$.

If the symmetry is **continuous** (i.e. it depends on a parameter we can make small), then

$$0 = \frac{\delta \mathcal{L}}{\delta \alpha} = \sum_n \left\{ \underbrace{\left[\frac{\partial \mathcal{L}}{\partial \phi_n} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \right]}_{\text{0 from Euler-Lagrange}} \frac{\delta \phi_n}{\alpha} + \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \frac{\delta \phi_n}{\delta \alpha} \right] \right\}$$

We simply have that $\partial_\mu J_\mu = 0$, where J_μ is called the **Noether current.**

$$\boxed{J_\mu = \sum_n \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \frac{\delta \phi_n}{\delta \alpha}}$$

A vector field J_μ that satisfies $\partial_\mu J_\mu = 0$ is also called a **conserved current** because the associated **charge** Q is conserved.

$$Q \equiv \int d^3x \; J_0$$
$$\partial_t Q = \int d^3x \; \partial_t J_0 = \int d^3x \; \nabla \cdot \vec{J} = 0$$

Noether's theorem: For every continuous symmetry of the Lagrangian, there exists an associated current that's conserved when the equations of motion (Euler-Lagrange) are satisfied.

- The symmetry must be **continuous,** not discrete.
- The current must be **on-shell,** i.e. equation of motion satisfied.
- Works for **global symmetries** α, not local symmetries $\alpha(x)$.

3.3 Spacetime Energy-Momentum Tensor

Spacetime translation invariance: physics at x^ν is the same at $y^\nu = x^\nu - \xi^\nu$ where ξ^ν is a constant. For small ξ^ν,

$$\phi(x) \to \phi(x + \xi) \approx \phi(x) + \xi^\nu \partial_\nu \phi(x)$$
$$\frac{\delta \phi}{\delta \xi^\nu} = \partial_\nu \phi$$

Then if we take the variational derivative of the Lagrangian,

$$\frac{\delta \mathcal{L}}{\delta \xi^\nu} = \partial_\mu \left(\sum_n \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \frac{\delta \phi_n}{\delta \xi^\nu} \right)$$
$$\partial_\nu \mathcal{L} = \partial_\mu \left(\sum_n \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi \right)$$
$$0 = \partial_\mu \left(\sum_n \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi - g_{\mu\nu} \mathcal{L} \right)$$

Now we may define the **energy-momentum tensor** as this current such that $\partial_\mu \mathcal{T}_{\mu\nu} = 0$.

$$\boxed{\mathcal{T}_{\mu\nu} = \sum_n \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_n)} \partial_\nu \phi_n - g_{\mu\nu} \mathcal{L}}$$

The conserved charges are:

$$Q_\nu = \int d^3x \mathcal{T}_{\mu\nu}$$

They correspond to the total energy and momentum of the system, which are of course conserved.

3.4 Concrete Example 1: Coulomb Interaction

This example will show common substitutions $\Delta \leftrightarrow -\vec{k}^2$ and $\Box \leftrightarrow -k^2$ in field theory.

3.4.1 Setup

Assume a static charge e at the origin.

$$\text{4-current: } J_\mu(x) = (\rho(x), \vec{0}) = (e\delta^3(x), \vec{0})$$
$$\text{Lagrangian: } \mathcal{L} = -\frac{1}{4}F_{\mu\nu}^2 - A_\mu J_\mu \ , \quad F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu$$

From Euler-Lagrange, we get

$$\frac{\partial \mathcal{L}}{\partial A_\nu} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} = 0 \ \longrightarrow \ -J_\nu - \partial_\mu(-\partial_\mu A_\nu) - \partial_\nu(\partial_\mu A_\mu) = 0$$
$$\partial_\mu F_{\mu\nu} = J_\nu$$

This is just Maxwell's equations in the presence of a source. We could rewrite further as:

$$\partial_\mu(\partial_\mu A_\nu - \partial_\nu A_\mu) = J_\nu$$
$$\Box A_\nu - \partial_\nu(\partial_\mu A_\mu) = J_\nu$$
$$\Box A_\nu = J_\nu \qquad \text{(Lorenz gauge } \partial_\mu A_\mu = 0\text{)}$$

Another way to write it (common in field theory) is as a propagator:

$$\boxed{A_\nu(x) = \frac{1}{\Box} J_\nu(x) \ , \quad \Pi_A \equiv \frac{1}{\Box}}$$

- A_ν is a field determined by source J_ν after it **propagates** with propagator Π_A.
- This representation will be useful for Feynman diagrams later on.

3.4.2 Fourier Transform

Note the following common substitutions in field theory:

$$\Delta^n \delta^3(\vec{x}) = \int \frac{d^3k}{(2\pi)^3} \Delta^n e^{i\vec{k}\vec{x}} = \int \frac{d^3k}{(2\pi)^3} (-\vec{k}^2)^n e^{i\vec{k}\vec{x}}$$
$$\Box^n \delta^4(x) = \int \frac{d^4k}{(2\pi)^4} \Box^n e^{ik_\mu x^\mu} = \int \frac{d^4k}{(2\pi)^4} (-k^2)^n e^{ik_\mu x^\mu}$$
$$\Delta \leftrightarrow -\vec{k}^2 \ , \quad \Box \leftrightarrow -k^2$$

In our Coulomb interaction problem, which is time-invariant,

$$A_0(x) = \frac{e}{\Box}\delta^3(x) = -\frac{e}{\Delta}\delta^3(x)$$

Now we take the fourier transform to go to the frequency space:

$$\begin{aligned} A_0(x) = \int \frac{d^3k}{(2\pi)^3} \frac{e}{\vec{k}^2} e^{i\vec{k}\vec{x}} &= \frac{e}{(2\pi)^3} \int_0^\infty k^2 dk \int_{-1}^1 d\cos\theta \int_0^{2\pi} d\phi \ \frac{1}{k^2} e^{ikr\cos\theta} \\ &= \frac{e}{(2\pi)^2} \int_0^\infty dk \ \frac{e^{ikr} - e^{-ikr}}{ikr} \\ &= \frac{1}{2} \frac{e}{(2\pi)^2} \frac{1}{ir} \int_{-\infty}^\infty dk \ \frac{e^{ikr} - e^{-ikr}}{k} \end{aligned}$$

Above is a compplex analysis problem. We take the limit as $\delta \to 0$ and separate residues.

$$\int_{-\infty}^{\infty} \frac{e^{ikr} - e^{-ikr}}{k} = \lim_{\delta\to 0} \int_{-\infty}^{\infty} dk \frac{e^{ikr} - e^{-ikr}}{k + i\delta}$$

$$= \lim_{\delta\to 0} \left[\underbrace{\int_{-\infty}^{\infty} dk \ \frac{e^{ikr}}{k + i\delta}}_{0} - \underbrace{\int_{-\infty}^{\infty} dk \ \frac{e^{-ikr}}{k + i\delta}}_{2\pi i(e^{-\delta r})} \right] = \lim_{\delta\to 0} \ 2\pi i e^{-\delta r} \to 2\pi i$$

Now we can recover the Coulomb interaction, up to constant factors:

$$\boxed{A_0(x) = \frac{1}{2} \frac{e}{(2\pi)^2} \frac{1}{ir} \cdot 2\pi i = \frac{1}{4\pi} \frac{e}{r}}$$

3.5 Concrete Example 2: Graviton

This example will highlight the Green's function method of expanding in small perturbations.

We start with the heuristic Lagrangian of the graviton. This isn't exactly correct, but just note that there are self-interaction terms.

$$\mathcal{L} = -\frac{1}{2} h \Box h + \frac{1}{3} \lambda h^3 + Jh \ \longrightarrow \ \frac{1}{2} (\partial_\mu h)(\partial^\mu h) + \frac{1}{3} \lambda h^3 + Jh$$

$h =$ gravitational potential

Our equation of motion is $\Box h - \lambda h^2 - J = 0$.

3.5.1 Green's Function Expansion

We can try to expand in λ for small λ.

1. If $\lambda = 0$, $h_0 = \frac{1}{\Box} J$.
2. Else, set $h = h_0 + h_1$ with $h_1 = O(\lambda)$.
$$\Box(h_0 + h_1) - \lambda(h_0 + h_1)^2 - J = 0$$
3. Solve for the perturbation up to $O(\lambda^2)$.
$$\Box h_1 = \lambda h_0^2 + O(\lambda^2)$$
$$h_1 \approx \lambda \frac{1}{\Box} h_0^2 = \lambda \frac{1}{\Box} \left[(h_0)(h_0) \right]$$
$$= \lambda \frac{1}{\Box} \left[\left(\frac{1}{\Box} J \right) \left(\frac{1}{\Box} J \right) \right]$$
4. Now we have our final answer to $O(\lambda^2)$, but we can go to higher powers using the same procedure. This is called the **Green's function method.**

$$\boxed{h \approx \frac{1}{\Box} J + \lambda \frac{1}{\Box} \left[\left(\frac{1}{\Box} J \right) \left(\frac{1}{\Box} J \right) \right] + O(\lambda^2)}$$

3.5.2 Propagator Expansion

This is similar to before, but specifically in terms of propagators.

1. Define $\Pi = -\frac{1}{\Box}$ as our solution to:

$$\Box_x \Pi(x,y) = -\delta^4(x-y) \ , \quad \Box_x \equiv g^{\mu\nu}\partial_\mu\partial_\nu$$

   Then we have

$$\Pi(x,y) = \int \frac{d^4k}{(2\pi)^4} e^{ik(x-y)} \frac{1}{k^2}$$

2. The field can be written as:

$$\begin{aligned} h(x) &= \int d^4y \ \delta^4(x-y)h(y) \\ &= -\int d^4y \ \left[\Box_y \Pi(x,y)\right] h(y) \\ &= -\int d^4y \ \Pi(x,y)\Box_y h(y) \end{aligned} \qquad \text{(with boundary conditions)}$$

3. From the free equation $\Box_y h_0(y) = J(y)$, we read off:

$$h_0(x) = -\int d^4y \ \Pi(x,y)J(y)$$

4. The next term in the expansion is $\Box h_1 \approx \lambda h_0^2$.

$$\Box_w h_1(w) = \lambda h_0^2(w) = \lambda \left(\int d^4y \ \Pi(w,y)J(y)\right)\left(\int d^4z \ \Pi(w,z)J(z)\right)$$

5. From the previous terms, we get that in total:

$$\boxed{\begin{gathered} h(x) \approx -\int d^4y \ \Pi(x,y)J(y) - \\ \lambda \int d^4w \left[\int d^4y \int d^4z \Pi(x,w)\Pi(w,y)\Pi(w,z)J(y)J(z)\right] + O(\lambda^2) \end{gathered}}$$

- Cf. previous method with $\Box$ directly embedded in the equation.
- In a sense, we've just recast that in terms of propagators.

3.6 Tangent: Old-Fashioned Perturbation Theory

OFPT was championed by Schwinger as an alternative to Feynman. Feynman diagrams are often easier, but OFPT shows the exact processes and "canceling" of infinities that are implicit in Feynman diagrams.

3.6.1 The Lippmann-Schwinger Equation

We're often interested in knowing the state of a system at time t_f, assuming we know it at time t_i. Assume the system has a fixed E.

$$\begin{aligned} \text{Time } t_i: &\quad H_0 |\phi\rangle = E |\phi\rangle \\ \text{Time } t_f: &\quad H |\psi\rangle = E |\psi\rangle \ , \quad H = H_0 + V \end{aligned}$$

- From t_i to t_f, $|\phi\rangle \to |\psi\rangle$.
- V is the perturbation.

From the above, we can recover the **Lippmann-Schwinger equation:**

$$\begin{aligned} H_0 |\phi\rangle - E |\phi\rangle = 0 &\longrightarrow (H_0 - E) |\phi\rangle = 0 \\ H |\psi\rangle - E |\psi\rangle = 0 &\longrightarrow (H_0 + V - E) |\psi\rangle = 0 \end{aligned}$$

$$\boxed{|\psi\rangle = |\phi\rangle + \frac{1}{E - H_0} |\psi\rangle}$$

- $\Pi_{LS} = 1/(E - H_0)$ is called the **Lippmann-Schwinger kernel.**
- To express $|\psi\rangle$ entirely in terms of $|\phi\rangle$, define the **transfer matrix** T such that $V |\psi\rangle = T |\phi\rangle$. Then

$$|\psi\rangle = \left(1 + \frac{1}{E - H_0} T\right) |\phi\rangle$$

3.6.2 Perturbative Expansion

1. Operate on both sides by V.

$$\underbrace{V |\psi\rangle}_{T|\phi\rangle} = \left(V + V \frac{1}{E - H_0} T\right) |\phi\rangle$$

2. Assuming that contracting with any $\langle \phi_j |$ satisfies the equation (i.e. in matrix form all entries are equal),

$$T = V + V \frac{1}{E - H_0} T$$

3. We can solve the above recursively.

$$\begin{aligned} T &= V + V \frac{1}{E - H_0} \left[V + V \frac{1}{E - H_0} \left(V + V \frac{1}{E - H_0} (\cdots)\right)\right] \\ &= V + V \frac{1}{E - H_0} V + V \frac{1}{E - H_0} V \frac{1}{E - H_0} V + \cdots \\ &= V + V \Pi_{LS} V + V \Pi_{LS} V \Pi_{LS} V + \cdots \end{aligned}$$

4. Contract by $\langle\phi_f|$ and $|\phi_i\rangle$ and let e.g. $T_{fi} = \langle\phi_f|T|\phi_i\rangle$. Inserting some resolutions of the identity, we come to the perturbative expansion of **old-fashioned perturbation theory.**

$$\boxed{\begin{gathered} T_{fi} = V_{fi} + \sum_j V_{fj}\Pi_{LS}(j)V_{ji} + \sum_{j,k} V_{fj}\Pi_{LS}(j)V_{jk}\Pi_{LS}(k)V_{ki} + \cdots \\ \Pi_{LS}(k) = \frac{1}{E - E_k} \end{gathered}}$$

Completely valid, and Schwinger was undoubtedly a genius. But physicists like geometry, and Feynman diagrams supplied that.

4 Cross Sections and Decay Rates

4.1 Cross Sections

In scattering experiments, we consider asymptotic states at $t = \pm\infty$ (roughly). Initial state $|i\rangle$, final state $|f\rangle$.

The scattering matrix or **S-matrix** is an operator that takes the initial state to the final state. Its elements are $S_{fi} \equiv \langle f|S|i\rangle$.

*Note: this is Heisenberg picture. Time evolution encoded in operators, not states.

4.1.1 in the Classical & Quantum Mechanical Picture

- Classically, the "cross sectional area" is

$$\sigma = \frac{[\#\text{ particles scattered}]}{[\text{time}] \times [\text{velocity}] \times [\text{beam number density}]} = \frac{1}{T}\frac{1}{\Phi}N$$

  We can also measure the differential cross section $d\sigma/d\Omega$, where $d\Omega$ is the angle. Tells us shape of object/potential.

- Quantum Mechanically, we generalize to the "cross section," an abstract measure of the interaction strength. There's a probability to scatter.

$$d\sigma = \frac{1}{T}\frac{1}{\Phi}dP\ , \quad P = \frac{N_{\text{scattered}}}{N_{\text{incident}}}$$

4.1.2 In Quantum Field Theory

Consider a 2-particle scattering, $\vec{p}_1 + \vec{p}_2 \to \{\vec{p}_j\}$. Now we must consider cases where particle number is not conserved.

1. in the CM frame, $\Phi = |\vec{v}_1 - \vec{v}_2|/V$.

$$d\sigma = \frac{V}{T}\frac{1}{|\vec{v}_1 - \vec{v}_2|}dP$$

2. The probability to scatter is

$$dP = \frac{|\langle f|S|i\rangle|^2}{\langle f|f\rangle\langle i|i\rangle} d\Pi\,, \qquad \underbrace{d\Pi = \prod_j V\frac{dp_j}{(2\pi)^3}}_{\text{region of final-state momenta}}$$

3. Mathematical interlude: from the delta function definition, we get

$$\delta^3(p) = \frac{1}{(2\pi)^3}\int d^3x\, e^{ipx} \;\longrightarrow\; \delta^3(0) = \frac{1}{(2\pi)^3}\int d^3x = \frac{V}{(2\pi)^3}$$
$$\delta^4(0) = \frac{TV}{(2\pi)^4}$$

   $T, V \to \infty$ later, but now we leave these infinities in the calculation. Further note that with the creation operator $a_k^\dagger|0\rangle = \frac{1}{\sqrt{2\omega_k}}|k\rangle$,

$$[a_p, a_q^\dagger] = (2\pi)^3\delta^3(p-q)$$
$$\langle p|p\rangle = (2\pi)^3(2\omega_p)\delta^3(0)$$

   Thus $\langle p|p\rangle = 2\omega_p V \to 2E_pV$ with $\hbar = 1$.

4. Let $|i\rangle = |p_1\rangle\otimes|p_2\rangle$ and $|f\rangle = \prod_j|p_j\rangle$. Then

$$\langle i|i\rangle = (2E_1V)(2E_2V)\,, \quad \langle f|f\rangle = \prod_j(2E_jV)$$

5. S must be unitary to preserve probabilities. Let $S = e^{i\mathcal{T}}$ where $\mathcal{T}$ is the "transfer matrix." Assume small deviation, so $S \approx 1 + i\mathcal{T}$. Insert delta functions to make initial & final momenta equal, and let $\mathcal{M}$ be the non-trivial elements.

$$\mathcal{T} = (2\pi)^4\underbrace{\delta^4\left(\sum p_i^\mu - \sum p_f^\mu\right)}_{\equiv\delta^4(\Sigma p)}\mathcal{M}$$

   Now the elements of S are:

$$\langle f|S|i\rangle = i(2\pi)^4\delta^4(\Sigma p)\langle f|\mathcal{M}|i\rangle$$
$$|\langle f|S|i\rangle|^2 = \underbrace{\delta^4(0)\delta^4(\Sigma p)}_{\text{From squared-}\delta}(2\pi)^8\underbrace{|\langle f|\mathcal{M}|i\rangle|^2}_{\equiv|\mathcal{M}|^2}$$
$$= \delta^4(\Sigma p)\,TV(2\pi)^4|\mathcal{M}|^2$$

6. The probability to scatter is:

$$dP = \frac{\delta^4(\Sigma p)\,TV(2\pi)^4}{(2E_1V)(2E_2V)\times\prod_j(2E_jV)}|\mathcal{M}|^2\times\prod_j V\frac{d^3p_j}{(2\pi)^3}$$
$$\equiv \frac{T}{V}\frac{1}{(2E_1)(2E_2)}|\mathcal{M}|^2 d\Pi_{LIPS}$$

   here $d\Pi_{LIPS}$ is the Lorentz-invariant momentum space.

7. Overall, in QFT, the cross section is:

$$\boxed{\begin{aligned} d\sigma &= \frac{1}{(2E_1)(2E_2)|\vec{v}_1 - \vec{v}_2|}|\mathcal{M}|^2 d\Pi_{LIPS} \\ d\Pi_{LIPS} &= (2\pi)^4 \delta^4\left(\Sigma p\right) \times \prod_j \frac{d^3 p_j}{(2\pi)^3} \frac{1}{2E_{p_j}} \end{aligned}}$$

It's instructive to see how the infinities in T, V have canceled. That was a doozy, but Feynman diagrams will streamline this process.

4.2 Decay Rates

Differential decay rate: probability that $\vec{p}_1$ turns into $\{\vec{p}_j\}$ over time T.

$$d\Gamma = \frac{1}{T} dP$$

This can be treated as a $1 \to n$ scattering. Use the same steps as for cross sections.

$$\begin{aligned} \langle i|i\rangle = 2E_1 V \ , \quad \langle f|f\rangle = \prod_j 2E_j V \\ S = I + i\mathcal{T} \ , \quad \mathcal{T} = (2\pi)^4 \delta^4(\Sigma p)\mathcal{M} \\ |\left\langle f|S|i\right\rangle|^2 = (2\pi)^8 \delta^4(0)\delta^4(\Sigma p)|\left\langle f|\mathcal{M}|i\right\rangle|^2 \end{aligned}$$

Thus:

$$\begin{aligned} dP = \frac{|\left\langle f|S|i\right\rangle|^2}{\langle f|f\rangle \langle i|i\rangle} d\Pi &= \frac{(2\pi)^8 \delta^4(0)\delta^4(\Sigma p)|\left\langle f|\mathcal{M}|i|^2\right\rangle}{(2E_1 V)\prod_j (2E_j V)} d\Pi \\ &= \frac{(2\pi)^4 \cdot TV\delta^4(\Sigma p)|\mathcal{M}|^2}{(2E_1 V)\prod_j (2E_j V)} d\Pi \\ &= \frac{T}{2E_1} \times \underbrace{(2\pi)^4 \delta^4(\Sigma p) \prod_j \frac{d^3 p_j}{(2\pi)^3} \frac{1}{2E_j}}_{d\Pi_{LIPS}} \end{aligned}$$

The decay rate is:

$$\boxed{d\Gamma = \frac{1}{T} dP = \frac{|\mathcal{M}|^2}{2E_1} d\Pi_{LIPS}}$$